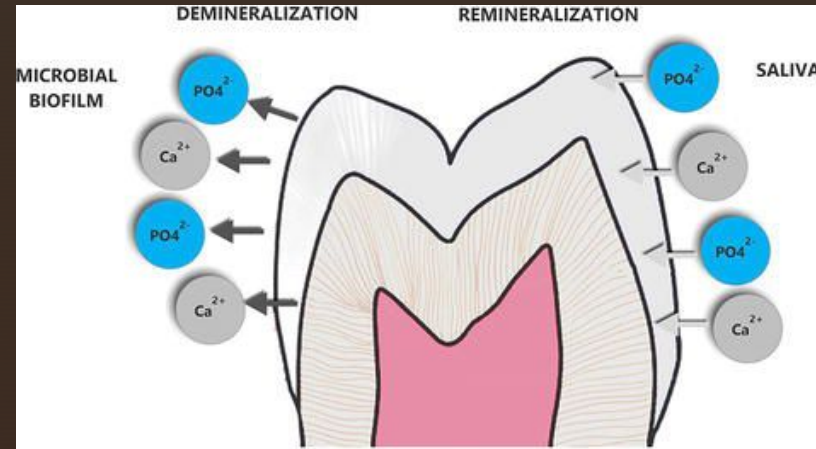
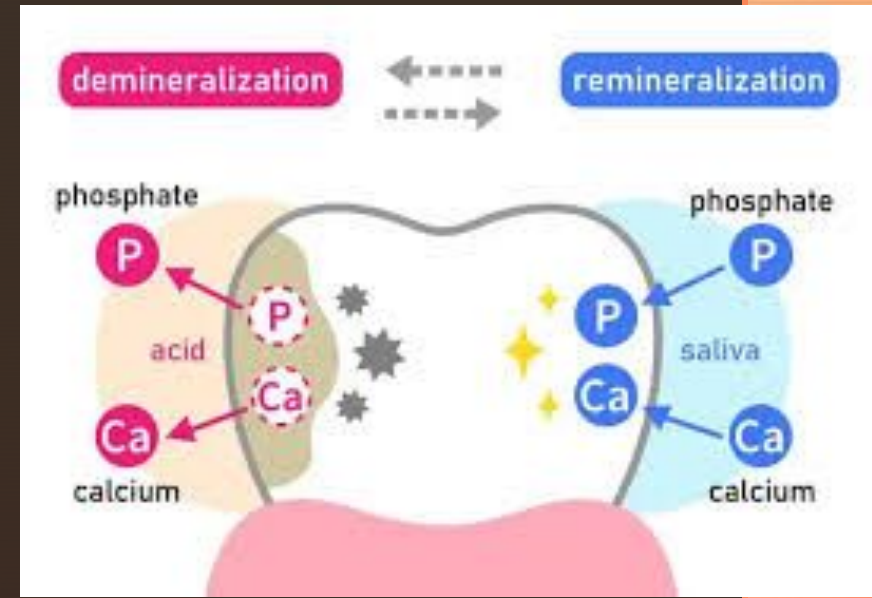




BY DR. WAJEEHA SHAMS ANSARI

Mineralization & Demineralization of Enamel and Dentin

INTRODUCTION



- Teeth are dynamic structures undergoing constant **mineral exchange**
- Balance between:
 - Mineralization (gain of minerals)**
 - Demineralization (loss of minerals)**
- Imbalance → **dental caries**

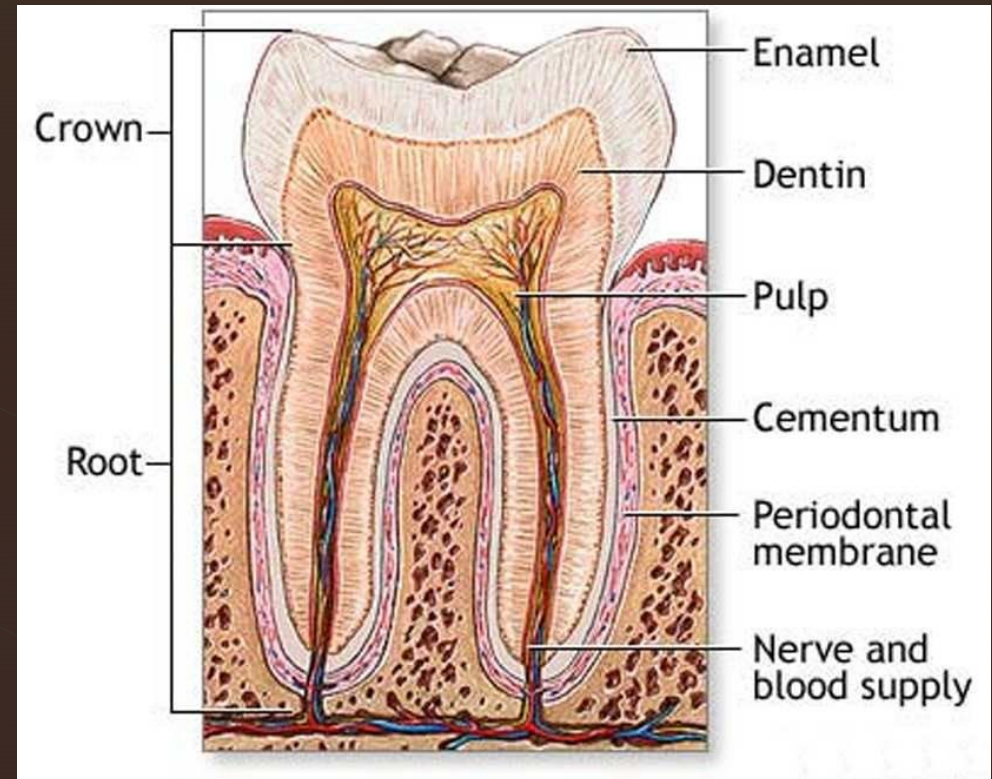
COMPOSITION OF TEETH

Enamel → outer protective layer

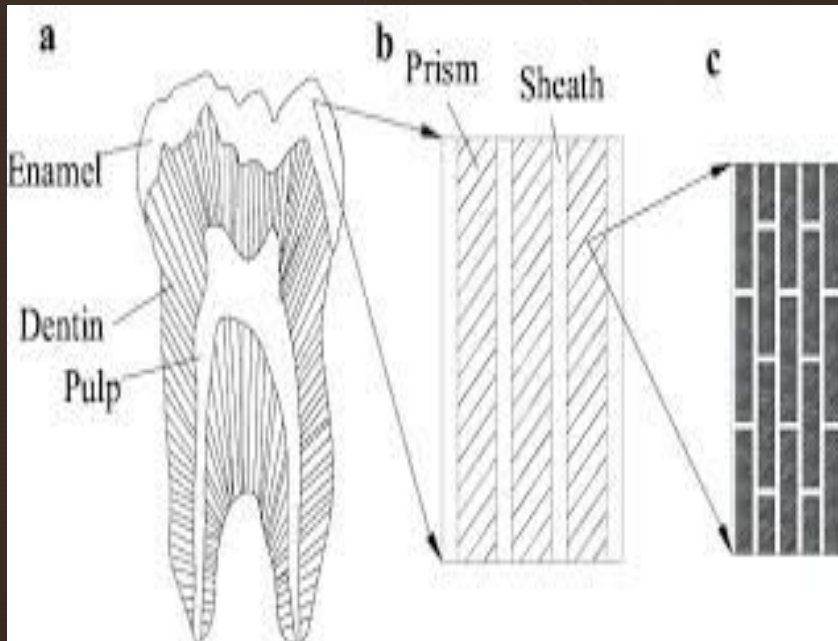
Dentin → underlying supportive tissue

Cementum → root covering

Pulp → vascular and nerve supply



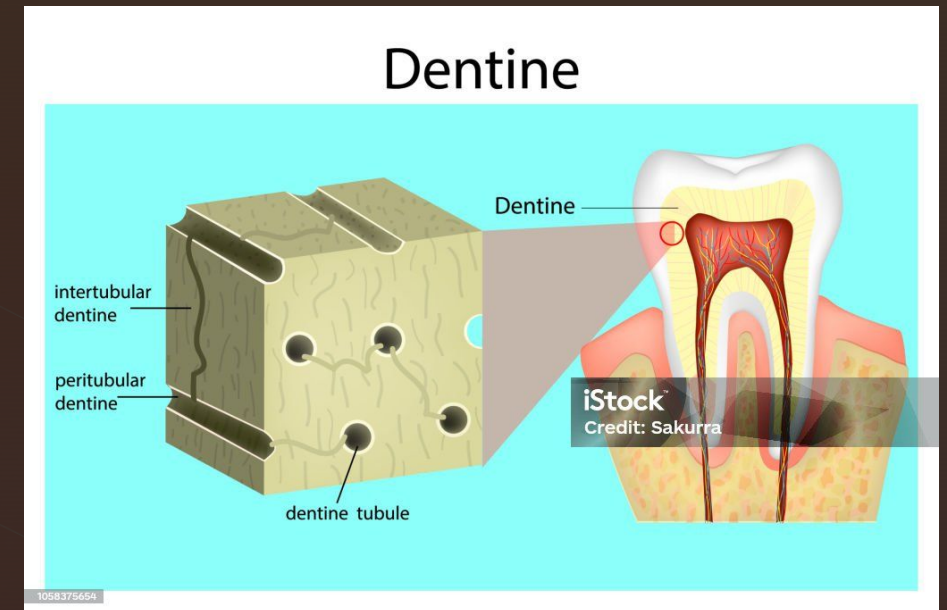
STRUCTURE OF ENAMEL



- ~96% inorganic (hydroxyapatite crystals)
- ~1% organic, ~3% water
- Organized into enamel rods (prisms)
- No cells → non-regenerative

STRUCTURE OF DENTIN

- ~70% inorganic, 20% organic, 10% water
- Organic matrix mainly **Type I collagen**
- Contains dentinal tubules
- Vital tissue with odontoblast processes

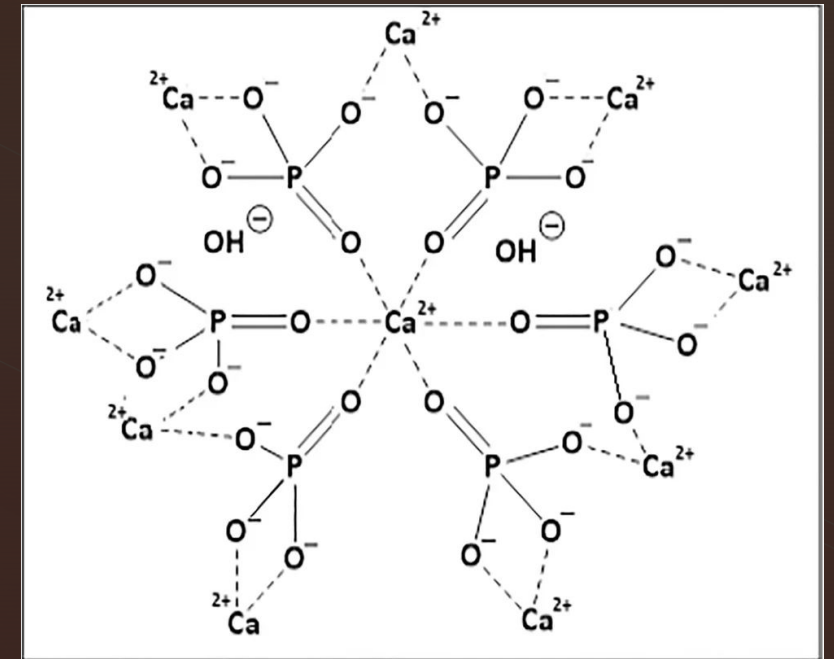


NORMAL AND CRITICAL pH

- Normal oral pH: ~6.7 – 7.4
- Critical pH of enamel: ~5.5
- Critical pH of dentin: ~6.2 – 6.5
- Below critical pH → demineralization begins

HYDROXYAPATITE CHEMISTRY

- Chemical formula: $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$
- Ionic equilibrium:
 - $\text{Ca}^{2+} + \text{PO}_4^{3-} \leftrightarrow \text{hydroxyapatite}$
- Stability depends on:
 - pH
 - Ion concentration
 - Presence of fluoride



What is Mineralization?



- Deposition of calcium and phosphate ions into organic matrix
- Formation of **crystalline hydroxyapatite**
- Occurs during:
 - Tooth development
 - Remineralization

MECHANISM OF MINERALIZATION

- Supersaturation of Ca^{2+} and PO_4^{3-}
- Nucleation (initial crystal formation)
- Crystal growth and maturation
- Regulation by matrix proteins

ROLE OF MATRIX PROTEINS

- Enamel:

Amelogenin

Enamelin

- Dentin:

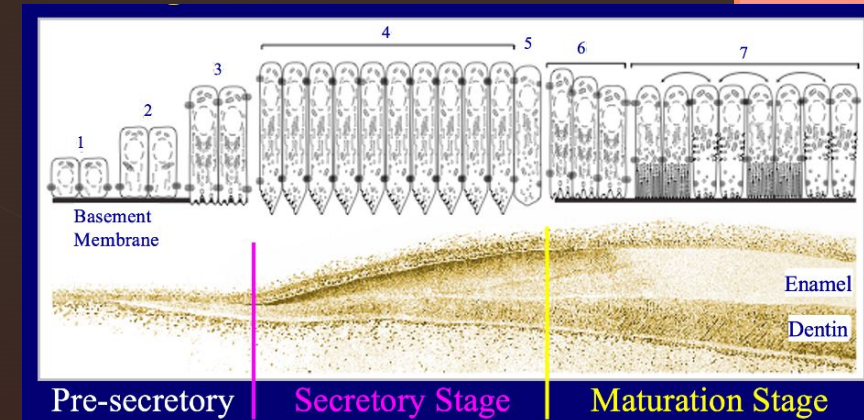
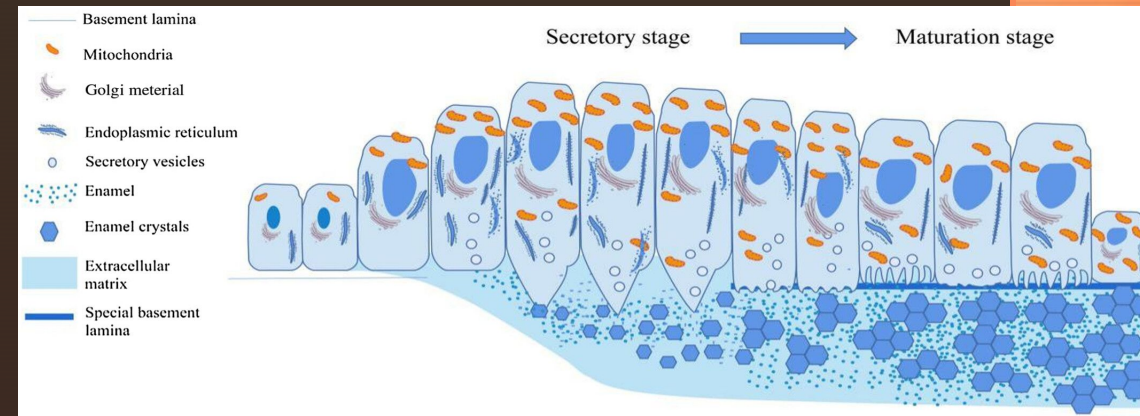
Collagen (Type I)

Dentin phosphoprotein

- Provide scaffold for mineral deposition

ENAMEL MINERALIZATION (AMELOGENESIS)

- Carried out by **ameloblasts**
- Two stages:
 - Secretory phase (matrix formation)
 - Maturation phase (mineral deposition)
- Results in highly organized crystals

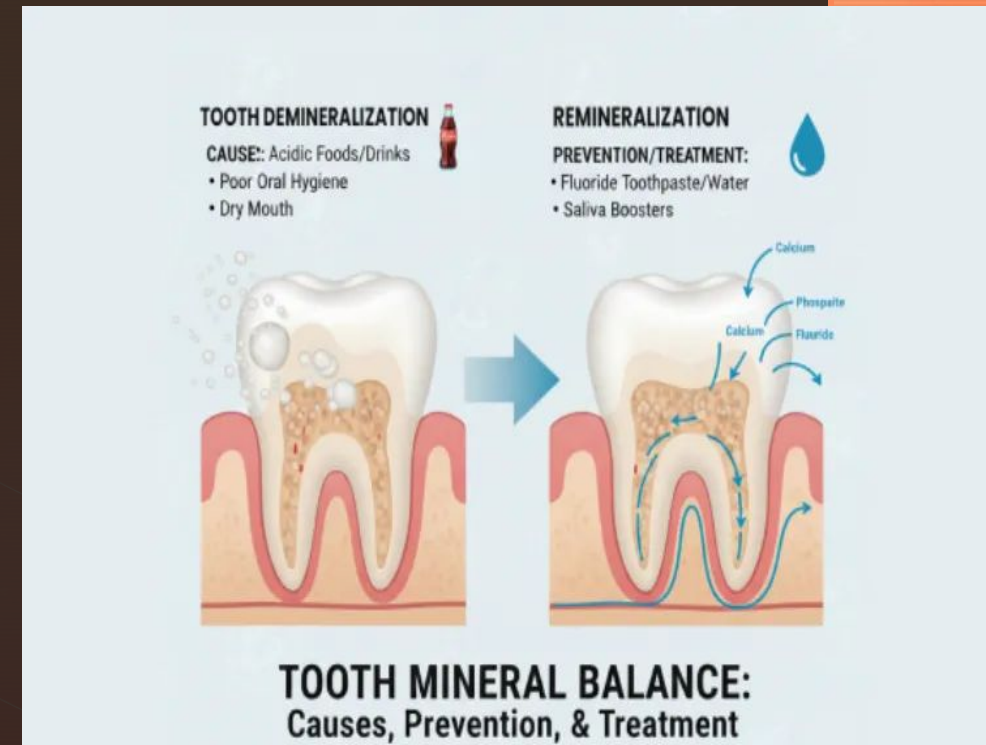


DENTIN MINERALIZATION (DENTINOGENESIS)

- Formed by **odontoblasts**
- Predentin (unmineralized) → mineralized dentin
- Mineralization occurs along collagen fibers

What is Demineralization?

- Dissolution of hydroxyapatite crystals
- Release of Ca^{2+} and PO_4^{3-} into saliva
- Triggered by **acidic environment**



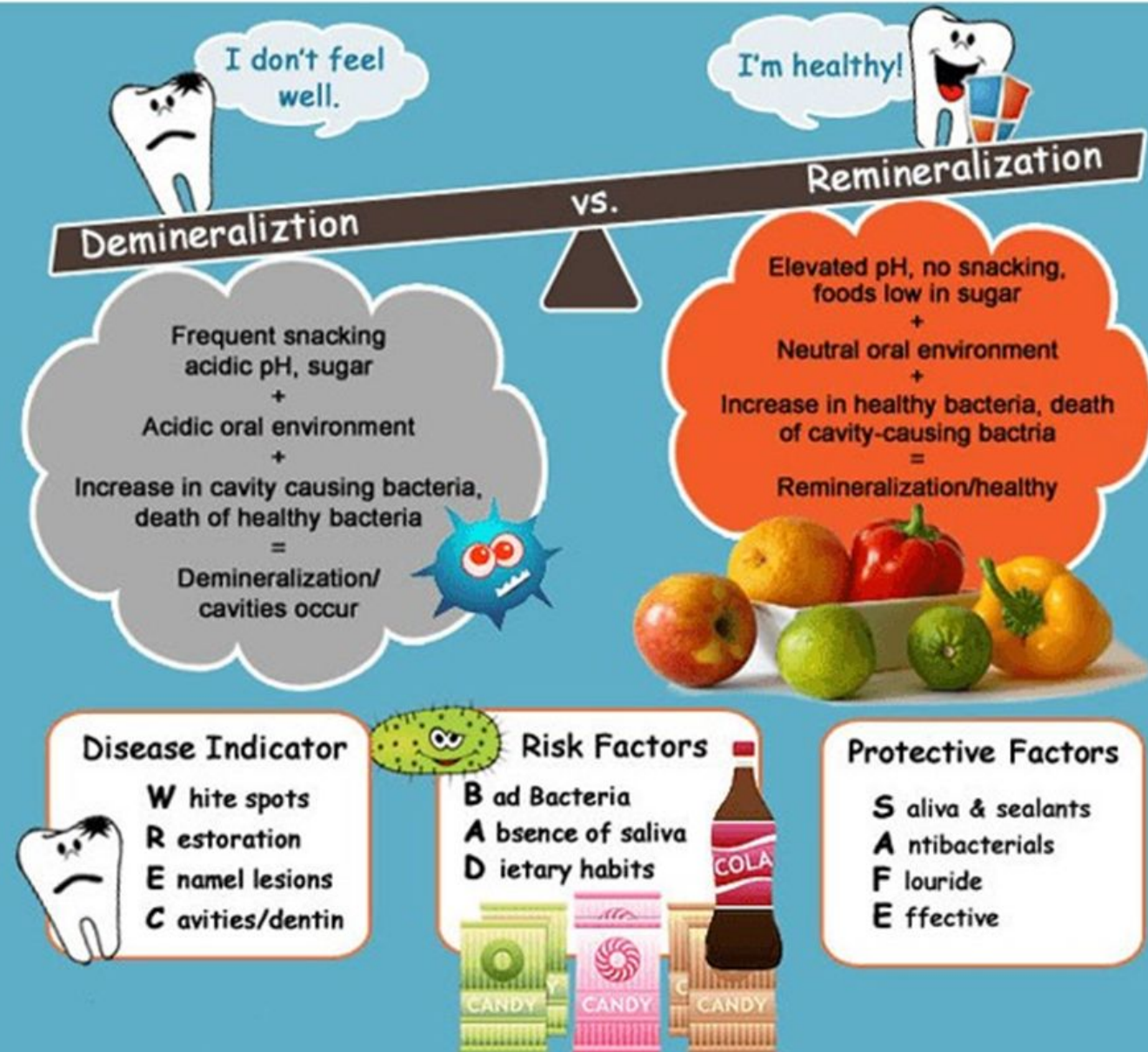
Causes of Demineralization

- Bacterial fermentation of carbohydrates
- Acid production (lactic acid)
- Common bacteria: ***Streptococcus mutans***
- Frequent sugar intake → prolonged low pH



Caries - means to 'rot' or 'decay'.

Dental Caries is the process of rotting away a tooth.



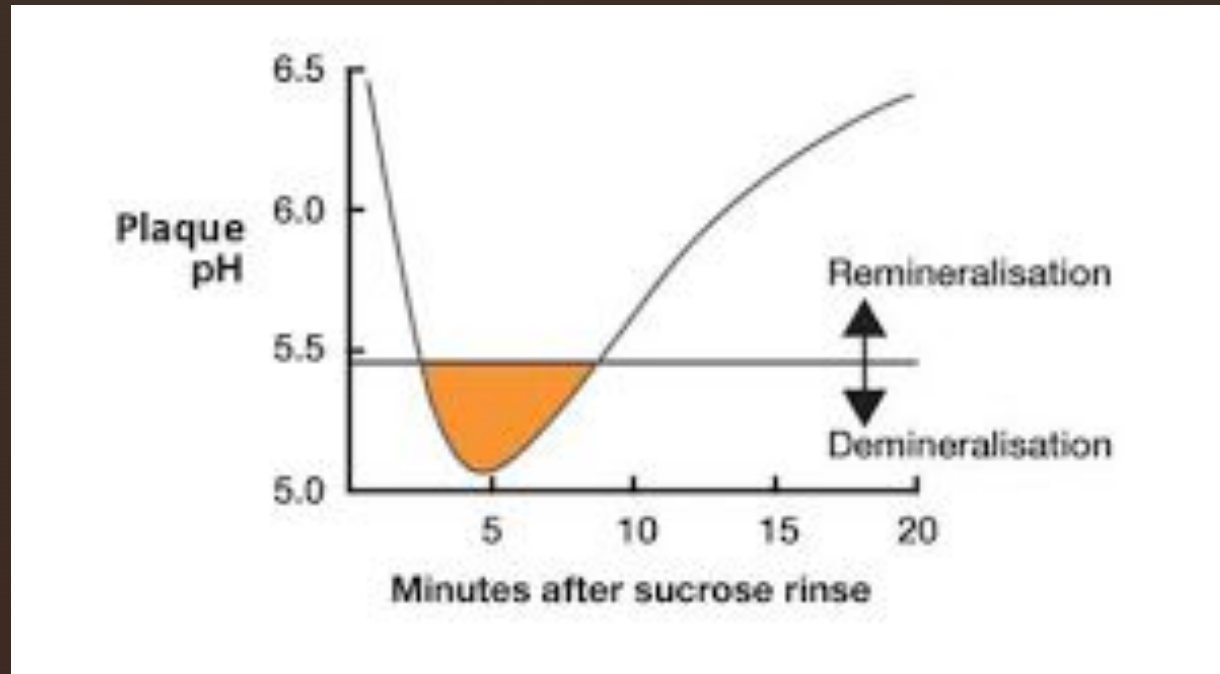
Mechanism of Demineralization

- Drop in pH below critical level
- Increased H^+ ions:

React with phosphate $\rightarrow$ reduces PO_4^{3-}

- Shifts equilibrium $\rightarrow$ **crystal dissolution**

STEPHAN CURVE



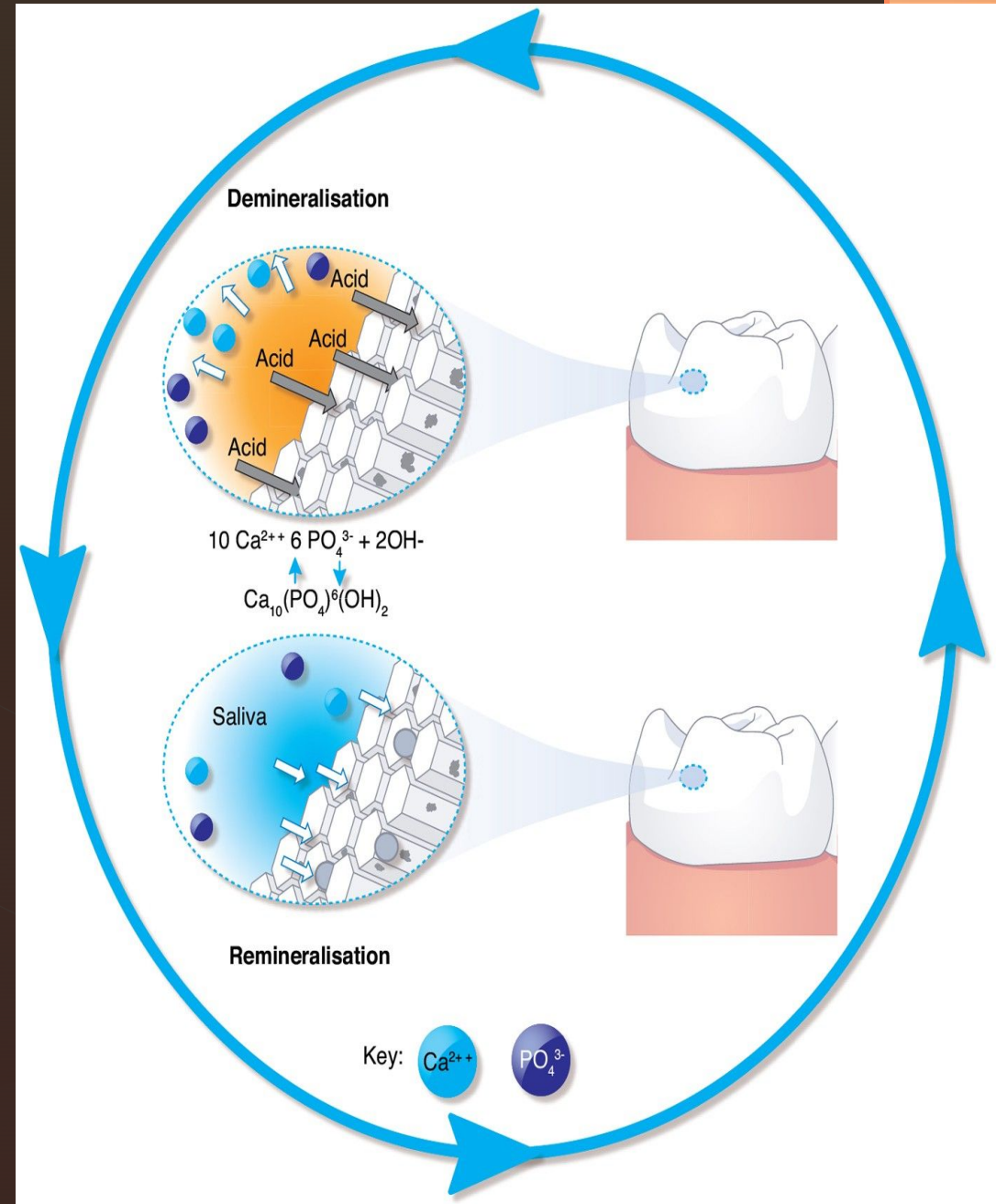
- Shows drop in plaque pH after sugar intake
- pH falls below critical level → demineralization
- Gradual recovery due to saliva

REMINERALIZATION

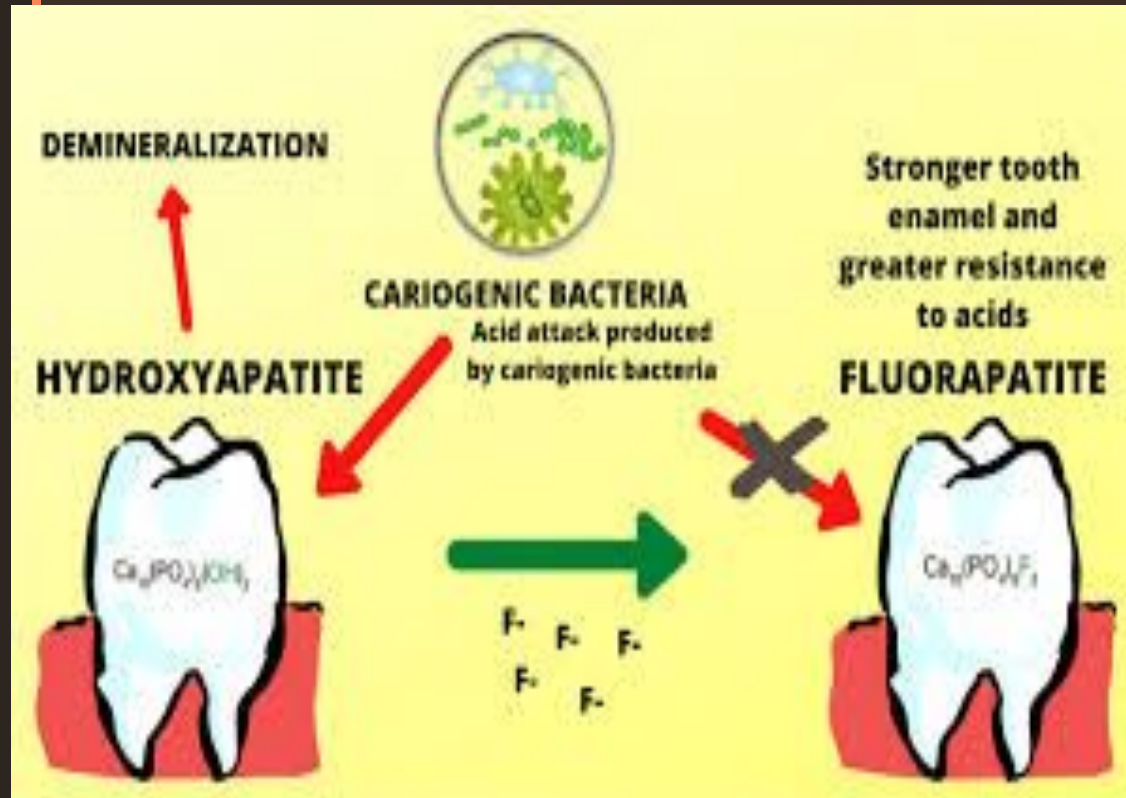
- Natural repair process
- Requires:
 - Calcium
 - Phosphate
 - Favorable pH
- Leads to restoration of enamel structure

ROLE OF SALIVA

- Buffering action (bicarbonate system)
- Maintains neutral pH
- Source of Ca^{2+} and PO_4^{3-}
- Contains antimicrobial proteins



ROLE OF FLUORIDE



- Converts hydroxyapatite → **fluorapatite**
- Lowers critical pH (~4.5)
- Enhances remineralization
- Inhibits bacterial enzymes

ENAMEL VS DENTIN

Feature	Enamel	Dentin
Mineral content	~96%	~70%
Organic matrix	Minimal	Collagen-rich
Critical pH	~5.5	~6.2–6.5
Sensitivity	No	Yes
Repair	No	Limited

CLINICAL CORRELATION

- Early lesion: **white spot lesion**
- Progression → cavitation
- Dentin demineralizes faster due to:
 - Higher organic content
 - Higher critical pH

PREVENTION

- Fluoride toothpaste
- Reduced sugar frequency
- Good oral hygiene
- Saliva stimulation (chewing gum)
- Regular dental visits

Thanks!

